

Artificial Intelligence - MCQ - Slot C

Total points 19/20 ?

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MCQ

19 of 20 points

C03-AT2

✓ Two logical sentences α and β are logically equivalent ($\alpha \equiv \beta$) if and only if: *1/1

- ☐ $\alpha \models \beta$
- ☐ $\alpha \Rightarrow \beta$ is a tautology
- ☒ $\alpha \models \beta$ and $\beta \models \alpha$
- ☐ $\alpha \wedge \beta$ is satisfiable



- ✓ Propositional resolution is **sound** and **refutation complete**. What does "refutation completeness" mean? *1/1
- ☐ Resolution can derive every sentence that is entailed by the KB.
 - ☒ If a set of clauses is unsatisfiable, resolution will always be able to derive a contradiction (the empty clause $\emptyset$). ✓
 - ☐ Resolution runs in polynomial time for arbitrary propositional formulas.
 - ☐ Resolution never generates redundant clauses.

- ✓ When Skolemizing a sentence where an existential quantifier lies within the scope of universal quantifiers (e.g., $\forall x \exists y \text{HasChild}(x, y)$), the existential variable y must be replaced by: *1/1
- ☐ A unique Skolem constant (e.g., C_1)
 - ☒ A Skolem function of the universally quantified variable (e.g., $f(x)$) ✓
 - ☐ A free variable y
 - ☐ The variable x

- ✓ Which of the following properties distinguishes a **logical agent** using Knowledge-Based Reasoning from a simple reflex agent? *1/1
- ☐ Logical agents map precepts directly to actions via lookup tables.
 - ☒ Logical agents maintain an internal state (Knowledge Base) represented in formal logic to deduce hidden aspects of the world. ✓
 - ☐ Logical agents only use probabilistic models to make decisions.
 - ☐ Logical agents do not require a domain-specific inference engine.



- ✓ Why does unification fail for the terms $p = \text{Knows}(\text{John}, x)$ and $q = \text{Knows}(x, \text{Jack})$? *1/1
- ☐ John and Jack are both functions.
 - ☒ Variable x cannot simultaneously bind to two different constants (John and Jack). ✓
 - ☐ The terms have different predicate arities.
 - ☐ Predicates cannot take variables as arguments.

- ✓ Given two terms $p = \text{Knows}(\text{John}, x)$ and $q = \text{Knows}(y, \text{Mother}(y))$, what is their Most General Unifier (MGU)? *1/1
- ☒ $\{x/\text{Mother}(\text{John}), y/\text{John}\}$ ✓
 - ☐ $\{x/\text{John}, y/\text{Mother}(y)\}$
 - ☐ $\{x/\text{Jane}, y/\text{John}\}$
 - ☐ Unification Fails

- ✓ The process of **Propositionalization** converts a FOL KB into Propositional Logic by creating instances of quantified sentences. What is a major theoretical limitation of propositionalization? *1/1
- ☐ It is unsound.
 - ☒ If the KB contains function symbols, the set of ground instances becomes infinite. ✓
 - ☐ It cannot handle implications.
 - ☐ It converts Horn clauses to non-Horn clauses.



✓ What is **Factoring** in First-Order Resolution, and why is it necessary for completeness? *1/1

- ☒ Factoring combines two unifiable literals within the same clause into a single literal using MGU θ . ✓
- ☐ Factoring splits a clause into two smaller clauses.
- ☐ Factoring removes all function symbols.
- ☐ Factoring converts disjunctions to implications.

✓ In the process of Knowledge Engineering for a domain, what step immediately follows **assembling the relevant knowledge**? *1/1

- ☐ Encoding general axioms about the domain.
- ☒ Deciding on a vocabulary of predicates, functions, and constants. ✓
- ☐ Positing specific queries to the inference engine.
- ☐ Skolemizing the KB.

✓ To prove that $KB \models \alpha$ using Resolution Refutation: * 1/1

- ☐ Add α to the KB and derive a tautology.
- ☒ Add $\neg\alpha$ to the KB, convert all clauses to CNF, and derive the empty clause ($\emptyset$). ✓
- ☐ Show that $KB \wedge \alpha$ is satisfiable.
- ☐ Show that $KB \Rightarrow \neg\alpha$ is true.



✓ Which of the following components are part of **First-Order Logic (FOL)** but are **NOT** present in Propositional Logic? *1/1

- ☐ Logical connectives ($\wedge, \vee, \neg$)
- ☒ Objects, Relations (Predicates), and Quantifiers ($\forall, \exists$)
- ☐ Truth values (True / False)
- ☐ Sentences and Parentheses



✓ What is the result of resolving Clause 1: $\neg P(x) \vee Q(x)$ with Clause 2: $P(\text{John}) \vee R(\text{Jane})$? *1/1

- ☒ $Q(\text{John}) \vee R(\text{Jane})$
- ☐ $\neg P(\text{John}) \vee R(\text{Jane})$
- ☐ $Q(x) \vee R(\text{Jane})$
- ☐ $\emptyset$ (Empty Clause)



✓ What is **Generalized Modus Ponens (GMP)**? * 1/1

- ☐ An inference rule for propositional resolution.
- ☒ A lifted inference rule that combines Unification with Modus Ponens directly on quantified Horn clauses. ✓
- ☐ A rule that eliminates existential quantifiers using Skolemization.
- ☐ An algorithm for converting FOL to CNF.



✗ Which of the following statements about **quantifier scope and order** in FOL is TRUE? *0/1

- ☐ $\forall x \exists y \text{ Loves}(x, y)$ is logically equivalent to $\exists y \forall x \text{ Loves}(x, y)$.
- ☒ Switching the order of two universal quantifiers ($\forall x \forall y$) changes the meaning of the statement. ✗
- ☐ $\exists y \forall x \text{ Loves}(x, y)$ implies $\forall x \exists y \text{ Loves}(x, y)$, but the converse is not generally true.
- ☐ Quantifiers can never be nested inside predicates.

Correct answer

- ☒ $\exists y \forall x \text{ Loves}(x, y)$ implies $\forall x \exists y \text{ Loves}(x, y)$, but the converse is not generally true.

✓ The First-Order Resolution Rule resolves two clauses: $l_i \vee \dots \vee l_k$ and $m_j \vee \dots \vee m_n$ if: *1/1

- ☐ l_i and m_j are identical.
- ☒ l_i and $\neg m_j$ are unifiable with MGU θ . ✓
- ☐ Both clauses contain only ground terms.
- ☐ Both clauses are Horn clauses.

✓ In Knowledge Engineering, what is the process of **Ontological Engineering**? *1/1

- ☐ Writing efficient Python algorithms for inference.
- ☒ Representing abstract general concepts—such as Actions, Time, Physical Objects, and Beliefs—in a structured hierarchy. ✓
- ☐ Converting all First-Order formulas to C code.
- ☐ Removing all quantifiers from a knowledge base.



✓ Consider the propositional sentence $\alpha = (P \vee Q) \wedge (\neg P \vee R)$. Under which truth assignment for (P, Q, R) does α evaluate to False? *1/1

- ☐ (T, T, T)
- ☐ (F, T, T)
- ☒ (T, F, F)
- ☐ (F, T, F)



✓ Which of the following sentences correctly represents "There is exactly one king" in FOL? *1/1

- ☐ $\exists x (\text{King}(x))$
- ☒ $\exists x (\text{King}(x) \wedge \forall y (\text{King}(y) \Rightarrow x = y))$
- ☐ $\forall x (\text{King}(x) \Rightarrow \exists y (x = y))$
- ☐ $\exists x \forall y (\text{King}(x) \vee \text{King}(y))$



✓ A propositional formula is in **Conjunctive Normal Form (CNF)** if it is: * 1/1

- ☒ A conjunction of clauses, where each clause is a disjunction of literals.
- ☐ A disjunction of clauses, where each clause is a conjunction of literals.
- ☐ A series of nested implication statements.
- ☐ A single negated implication.



✓ Consider the assertion: "No two people have the same social security number (SSN)." Select the correct FOL formulation: *1/1

- ☐ $\forall x \forall y (SSN(x) = SSN(y) \Rightarrow x = y)$
- ☐ $\forall x \forall y (x \neq y \Rightarrow SSN(x) \neq SSN(y))$
- ☒ Both A and B are correct.
- ☐ Neither A nor B is correct.



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